XUEWU CAI

Enterprise Applications & Workflow Delivery | Java, Python, AWS | Financial Analytics

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PROFESSIONAL SUMMARY

Financial engineering director and technology leader with more than 20 years of experience delivering enterprise financial platforms, quantitative analytics and trading systems. Leads modeling and forecasting delivery on AWS and IBM Symphony, with hands-on Java, Python and C/C++ expertise. Uses Copilot and other generative AI tools to streamline engineering workflows, while connecting financial requirements with practical software implementation.

CORE COMPETENCIES

Applications & Data: Java, Python, C/C++, MATLAB, SQL, Oracle, SQL Server, Spring Boot, Docker

Platforms & Integration: AWS, IBM Symphony, database connectivity, time-series infrastructure, market data feeds

Business Delivery: enterprise forecasting, financial model implementation, multi-project delivery, technical leadership

Finance & AI Tools: CRT, CECL, capital planning, DFAST; Copilot, Amazon Q, harness engineering

PROFESSIONAL EXPERIENCE

Director, Financial Engineering — Freddie Mac

May 2014 – Present

McLean, VA | Credit Risk Transfer (STACR / ACIS)

- Direct development of modeling tools combining prepayment/default models, AWS and IBM Symphony computing, and Intex valuation analytics to assess bond performance and transaction economics.
- Own an enterprise forecasting platform for income statements, balance sheets and CECL estimates, supporting business planning, capital planning and DFAST stress testing.
- Deliver RPL and synthetic-structure deal memorandum, capital, economics and ROE analytics; use Copilot and harness engineering to support delivery.
- Lead engineering projects for STACR bond issuance and ACIS reinsurance within a CRT program protecting more than $1.6 trillion in mortgage loans.
- Manage the LIBOR-to-SOFR transition within CRT modeling tools and analytics.

Campbell & Company | Research Engineer and Engineering Roles

Mar 2003 – Apr 2014

Washington, DC–Baltimore Area

Research Engineer, Portfolio and Risk Team (2009–2014); Quantitative Finance Engineer (2006–2008); Senior Software Engineer (2003–2006).

- Led architecture and development of a multithreaded, fault-tolerant Java/Portware/FIX trading platform supporting concurrent strategies, orders and multiple asset classes; supported more than $3 billion in capital and helped save tens of millions in slippage costs.
- Built a Java-based intraday cash equity trading system on Caché and market data/time-series infrastructure in MFC/C++; unified MATLAB access to Oracle, SQL Server, Caché and QADirect through a Java connectivity layer.
- Designed risk reporting for Investment Committee decisions and produced SEC, CFTC and OPERA reporting using MATLAB, Java/StyleReport and Panopticon.

Senior Software Engineer — Aquilent

Jun 1999 – Feb 2003

Java/Linux integration and data systems

- Built Java/Linux sensor integration systems with embedded Linux, JNI, serial communications and dynamic sensor discovery; developed Apache/Tomcat visualization portals and server-side satellite data components.

EDUCATION & PROFESSIONAL DEVELOPMENT

Master's Degree, Computer Science — University of Maryland Baltimore County

M.S., Computer Science & Engineering — Shanghai Jiao Tong University

Bachelor's Degree, Applied Mathematics — Dalian University of Technology

Professional development: CAIA (Chartered Alternative Investment Analyst) Level I

LANGUAGES

Chinese: native or bilingual proficiency | English: full professional proficiency